MNC Growth and Workforce Analytics Using Python

Project Documentation

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab / Jupyter Notebook

1. Project Overview

This project focuses on analyzing the Inc 5000 Companies dataset using Python to extract meaningful insights about company growth and workforce performance. The analysis includes data loading, data cleaning, data transformation, feature engineering, exploratory data analysis (EDA), statistical analysis, and visualization.

The project aims to understand company revenue growth, industry performance, workforce expansion, geographical distribution, revenue efficiency, and company age patterns. The analysis transforms raw company information into useful findings that can support business understanding and decision-making.

2. Data Source

Details	Project Information
Dataset Name	Inc 5000 Companies dataset
Original Source	Maven Analytics Data Playground
Source URL	https://mavenanalytics.io/data-playground
Domain	Finance
File Type	CSV
Number of Records	502 rows
Number of Original Attributes	14 fields
Dataset Coverage	Companies on the 2019 Inc 5000 list
Timeline	Founding years from 1909 to 2015
Geographical Coverage	43 states, 293 cities and 54 metropolitan-area labels

The dataset contains company profile information, including company name, location, industry, revenue, revenue growth, founding year, and workforce details.

3. Problem Statement

Organizations need to understand how companies perform across revenue, growth, workforce, industries, and geographical locations. However, raw company data may contain missing values, inconsistent formats, and information that requires transformation before meaningful analysis can be performed.

This project aims to clean and analyze the Inc 5000 Companies dataset to identify important business patterns, compare company and industry performance, understand workforce growth, and examine the relationships between revenue, growth, and company characteristics.

4. Objectives

- Analyze Company Growth — Identify companies with the highest revenue growth percentage.
- Compare Industry Performance — Determine which industries generate the highest total revenue and have thelargest workforce.
- Evaluate Workforce Growth — Identify companies with the highest employee growth compared to their previousworkforce.
- Analyze Geographical Distribution — Identify the states and cities with the highest number of companies andcompare business performance by location.
- Study Revenue and Growth Trends — Analyze the relationship between company revenue and revenue growthpercentage.
- Analyze Revenue per Employee — Identify companies with the highest revenue per employee to evaluate workforceproductivity.
- Study Company Age and Business Trends — Analyze the distribution of companies across founding years, comparethe growth of newer and older companies, and identify industries with the highest and lowest representation.

5. Attribute (Column / Feature) Details

Attribute Name	Description
Rank	Company rank in the Inc 5000 list.
Profile	Company profile reference or URL.
Company_name	Name of the company.
url	Company website or web address field.
state	State where the company is located.
city	City where the company is located.
Industry	Industry or business sector of the company.
Metro	Metropolitan-area label associated with the company.
Founded	Year in which the company was founded.

Revenue_in_millions	Company revenue converted to a numeric value in millions.
Revenue_Growth_Percentage	Company revenue growth percentage.
Years_On_List	Number of years the company appeared on the list.
Previous_Workers	Previous reported number of workers.
Current_Workers	Current reported number of workers.
Employee_Change	Difference between current and previous workers.
Employee_Growth_Percentage	Percentage change in workforce compared with previous workers.
Company_Size	Company category based on current workforce: Small, Medium or Large.
Company_Age	Company age calculated using 2015 as the reference year.
Revenue_Per_Employee	Revenue divided by current workforce, expressed as revenue per employee.

The original dataset contains 14 attributes. Additional calculated features were created during preprocessing and feature engineering to support the analysis.

6. Tools & Technologies

Technology	Usage
Python	Programming language used for data analysis.
Pandas	Loading, cleaning, transforming, grouping and analyzing tabular data.
NumPy	Numerical calculations and handling numeric values.
Matplotlib	Creating charts and visual representations.
Seaborn	Creating statistical charts such as count plots, box plots, scatter plots and heatmaps.
Jupyter Notebook / Google Colab	Writing and executing Python code and documenting the analysis.

7. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using data preprocessing and feature engineering techniques.

Data Cleaning Steps

- Checked the data types and structure of the dataset.
- Identified missing values in the columns.

- Replaced the text value “Unknown” in the Metro column with a missing value and filled those entries using the citycolumn.
- Checked duplicate records.
- Renamed columns for clearer and more consistent naming.
- Removed extra spaces from text columns.
- Converted the revenue column from text values such as “36.9 Million” into numeric values.
- Rounded revenue growth percentage values to two decimal places.
- Checked zero values in the previous and current workforce columns.

Feature Engineering

New features were created from existing columns to support additional analysis.

- Employee_Change: Current_Workers minus Previous_Workers.
- Employee_Growth_Percentage: Employee_Change divided by Previous_Workers, multiplied by 100 and rounded totwo decimal places.
- Company_Size: Small for fewer than 50 workers, Medium for 50 to 249 workers, and Large for 250 or more workers.
- Company_Age: Calculated as 2015 minus Founded, using 2015 as the reference year because the dataset periodends in 2015.
- Revenue_Per_Employee: Revenue in millions converted to the actual revenue amount and divided byCurrent_Workers.

A zero current-worker value was replaced with a missing value for the revenue-per-employee calculation to avoid division by zero. The resulting missing value was retained rather than being incorrectly changed to zero.

8. Analysis and Visualizations

The dataset was analyzed using Exploratory Data Analysis techniques, statistical measures, and visualizations to understand company growth, revenue, workforce patterns, industry performance, and geographical distribution.

- Initial dataset inspection and data quality checks.
- Descriptive statistical analysis of numerical variables.
- Univariate analysis to study individual variables.
- Bivariate analysis to examine relationships between two variables.
- Multivariate analysis to compare multiple business and workforce variables.
- Objective-wise analysis to answer the seven project objectives.

9. Exploratory Data Analysis (EDA)

Initial inspection and final EDA were performed to understand the structure, completeness, and statistical characteristics of the dataset.

Initial Inspection Included

- Verified the number of rows and columns.
- Reviewed column names and data types.
- Displayed sample records using head().
- Reviewed numerical summaries using describe().
- Checked missing values and duplicate records.

Final EDA Included

- Reviewed the cleaned and transformed dataset.
- Checked numerical columns and calculated features.
- Reviewed company-size distribution.
- Checked the missing revenue-per-employee value caused by a zero current-worker value.
- Prepared the dataset for statistical analysis and visualization.

The EDA helped confirm the dataset's structure and identify data characteristics that needed attention before business analysis.

10. Statistical Analysis

Statistical measures were used to summarize the dataset and understand the distribution and variability of company revenue, workforce, and revenue growth.

Measures Included

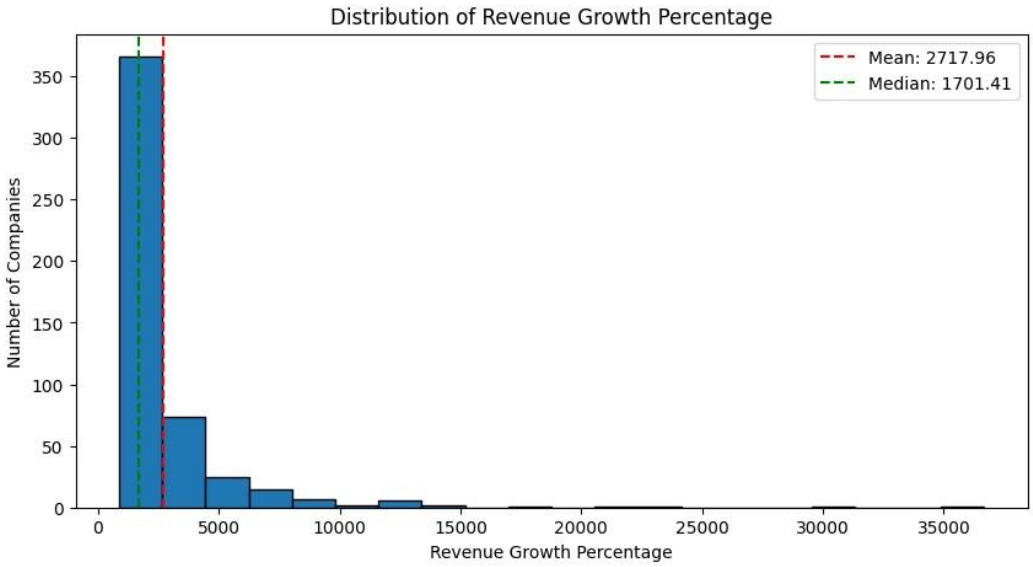
- Mean
- Median
- Mode
- Minimum value
- Maximum value
- Variance
- Standard deviation

11. Detailed Analysis and Visualizations

11.1 Statistical Analysis & Visualizations: Statistical measures summarize the centre and spread of company revenue growth and workforce performance. The following charts are placed under their corresponding statistical headings.

Central Tendency · Mean and Median

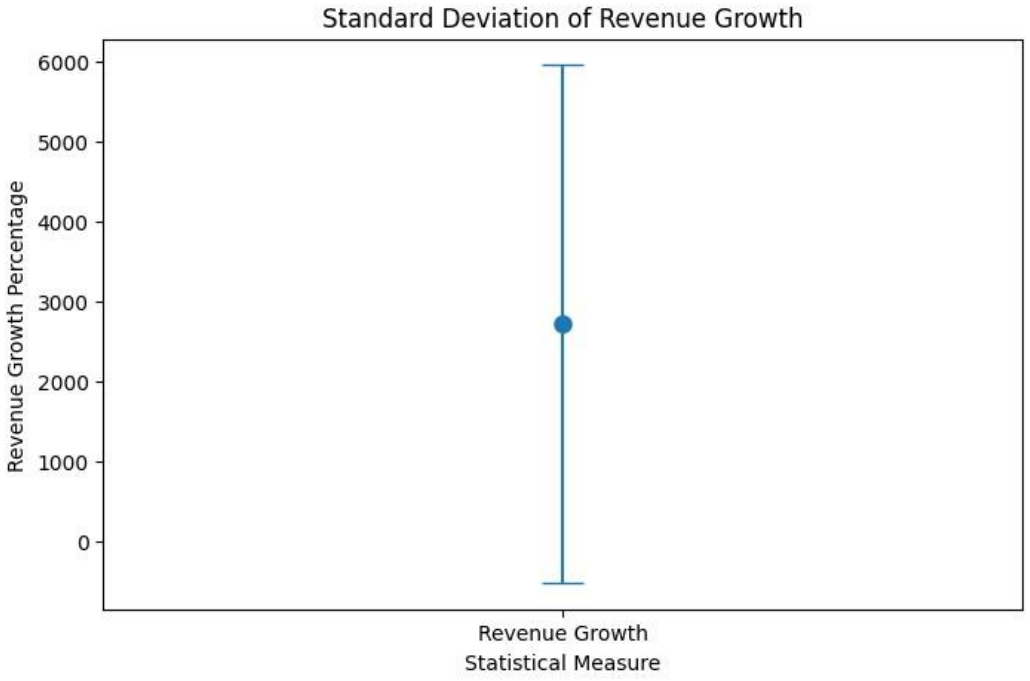
Mean and median were calculated for Revenue_Growth_Percentage. The notebook uses a histogram to show the revenue-growth distribution, with vertical lines marking the mean and median.



The chart compares average revenue growth with the middle value. The mean is higher than the median, suggesting that a small number of exceptionally high-growth companies pull the average upward. The median is useful for describing typical company growth when extreme values are present.

Variance and Standard Deviation

Variance measures how widely revenue-growth values are spread around their mean, while standard deviation expresses the spread in the same units as the growth percentage.



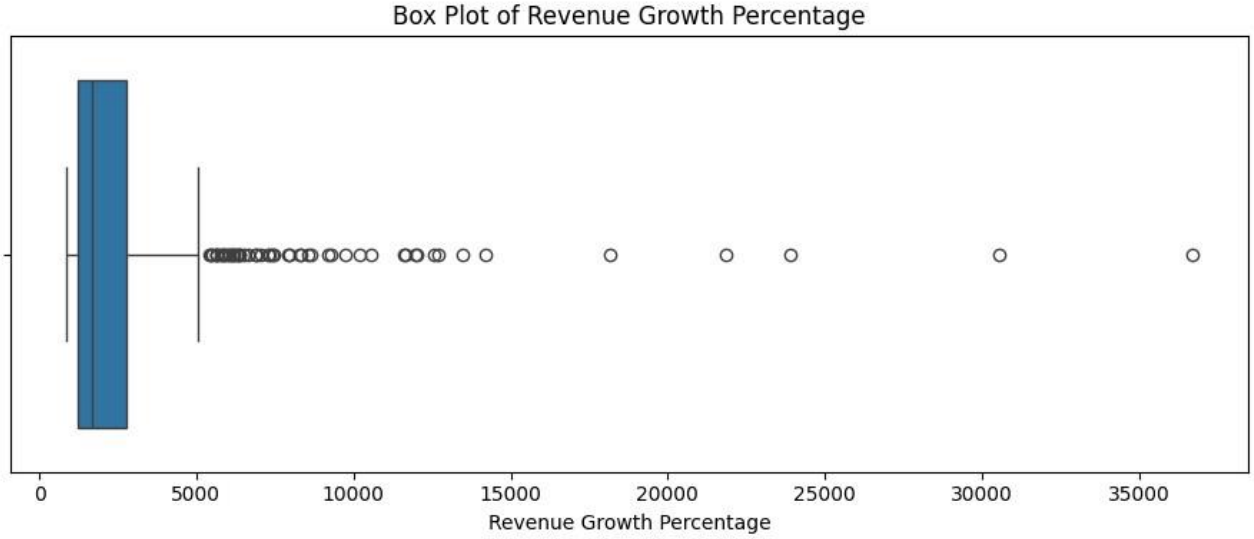
Revenue growth shows substantial variation across the 502 companies. The notebook reports a standard deviation of approximately 3,245.59 percentage points, larger than the mean of approximately 2,717.96%. This indicates considerable differences in growth rates across companies.

Skewness, Kurtosis and Outlier Detection

Skewness and kurtosis describe the shape of the revenue-growth distribution. The notebook reports skewness of approximately 5.40 and excess kurtosis of approximately 41.13.

Box Plot of Revenue Growth Percentage

The box plot is used to identify potential outliers in revenue growth.



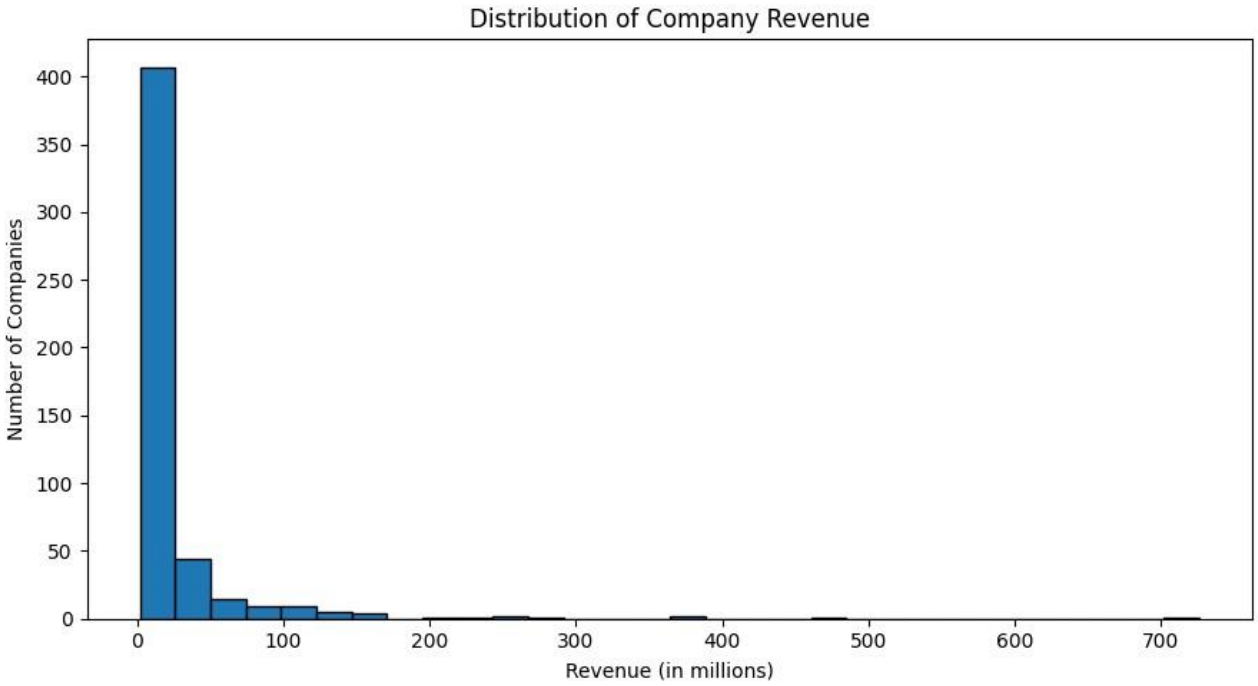
Interpretation: The notebook identifies 52 potential outliers using the 1.5 × IQR rule. These observations should be investigated rather than automatically removed, as they may represent genuine high-growth companies or data-quality issues.

11.2 Univariate Analysis

Univariate analysis examines one variable at a time to understand its distribution, frequency and unusual values. Each chart is positioned directly below its own analysis.

1. Histogram — Distribution of Company Revenue

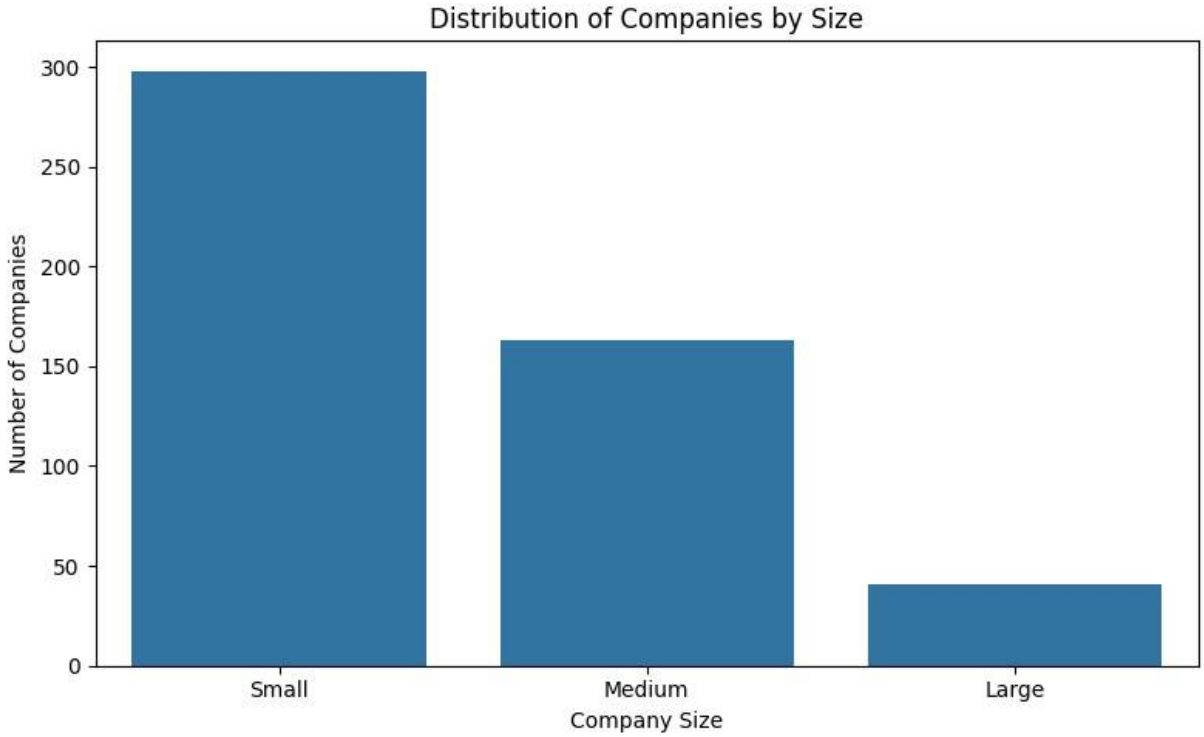
This histogram shows the distribution of company revenue across the 502 companies.



Interpretation: Average revenue is approximately $24.37 million, while median revenue is $8.05 million. A relatively small number of high-revenue companies increase the average, so the median is also useful when describing typical company revenue.

2. Count Plot — Distribution of Company Size

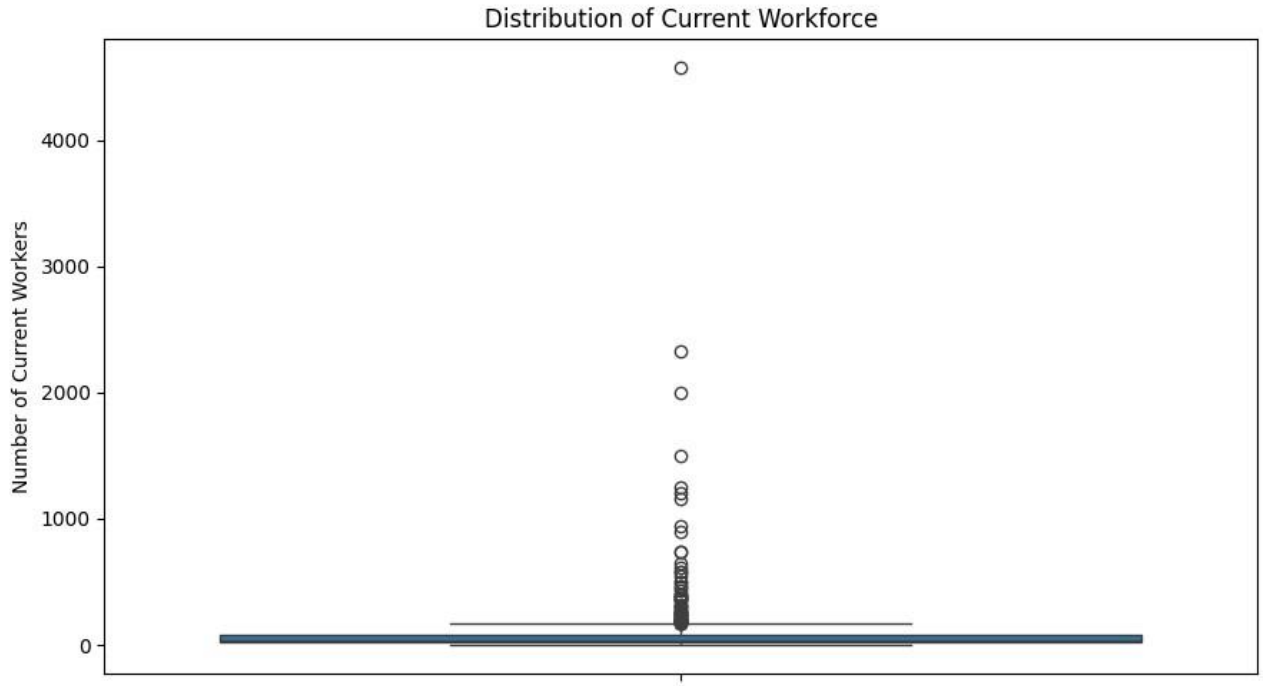
This count plot compares the number of Small, Medium and Large companies.



Interpretation: The dataset contains 298 Small companies, 163 Medium companies and 41 Large companies. Small companies make up the largest group, approximately 59% of the dataset.

3. Box Plot — Distribution of Workforce Size

This box plot shows the spread of current employee counts and potential workforce outliers.



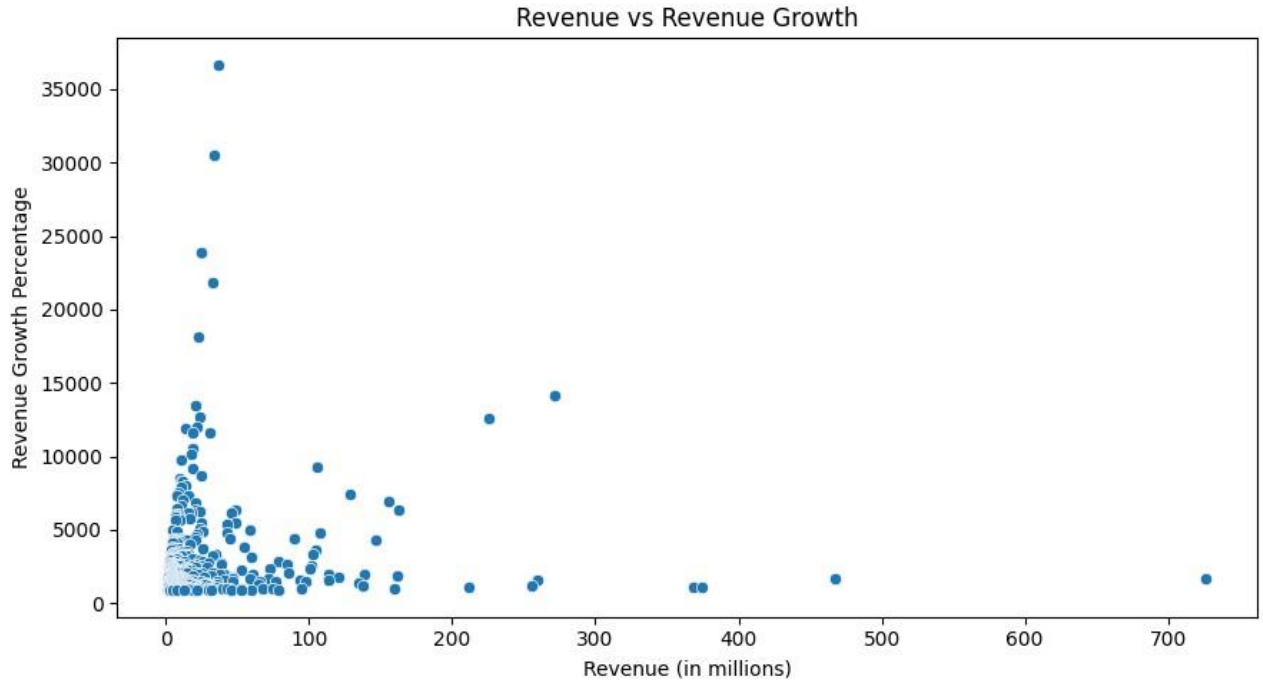
Interpretation: The median workforce is approximately 38 employees, compared with an average of approximately 104. A smaller number of companies with very large workforces increase the average.

11.3 Bivariate Analysis

Bivariate analysis studies the relationship between two variables to identify associations and compare performance.

1. Scatter Plot — Revenue vs Revenue Growth

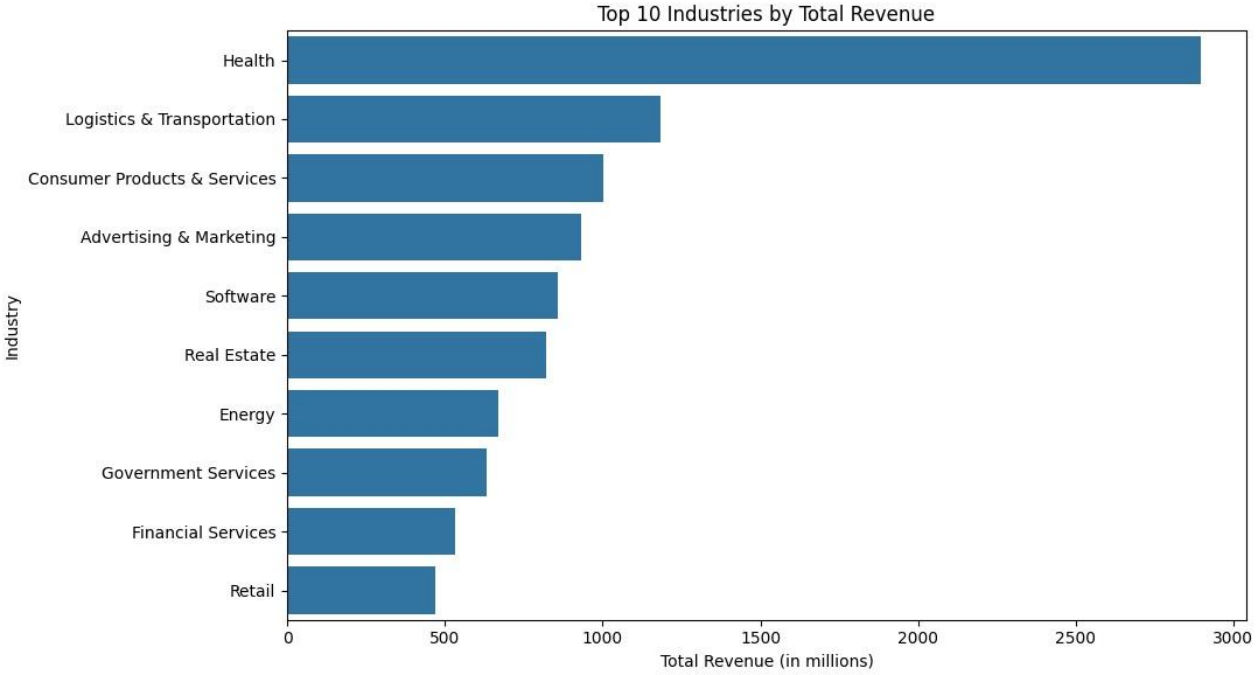
This chart examines whether companies with higher revenue also tend to have higher revenue growth.



Interpretation: The correlation is approximately 0.07, indicating a very weak positive relationship. Higher revenue does not necessarily mean higher revenue growth.

2. Bar Plot — Total Revenue by Industry

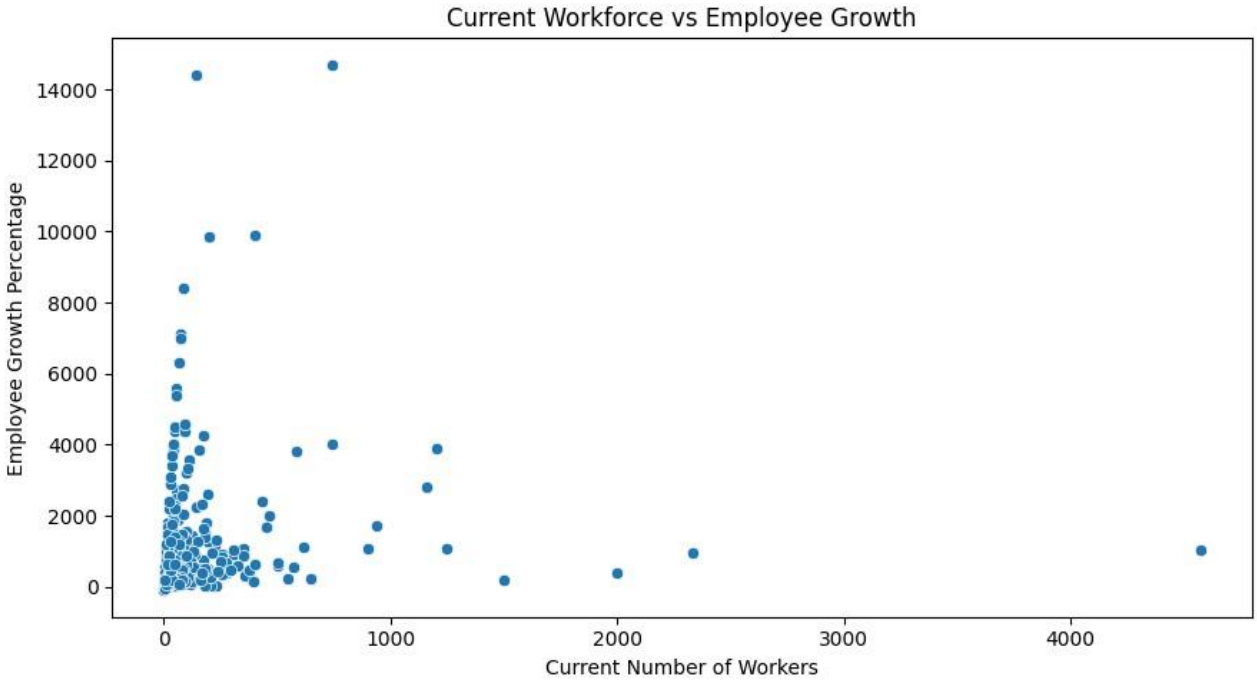
This chart identifies the ten industries contributing the highest combined revenue.



Interpretation: The Health industry generates the highest combined revenue at approximately $2,895.5 million, followed by Logistics & Transportation at approximately $1,183.8 million.

3. Scatter Plot — Current Workforce vs Employee Growth

This chart examines whether companies with larger workforces also tend to have higher employee growth percentages.



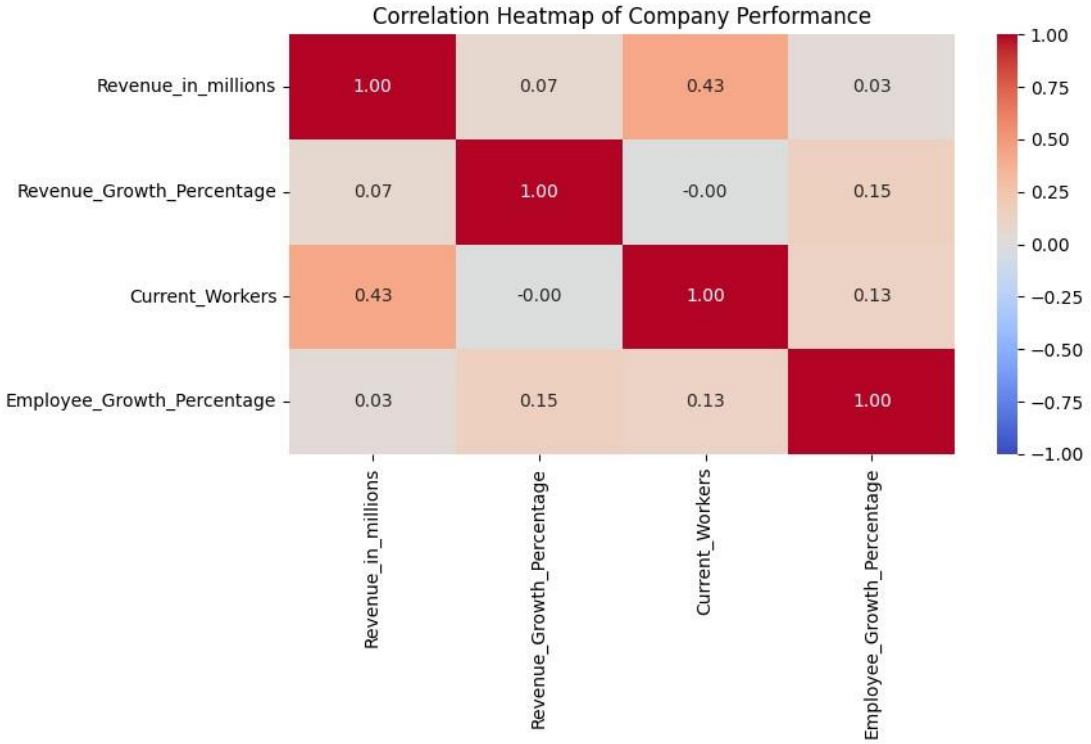
Interpretation: The correlation is approximately 0.03, indicating a very weak positive relationship. A larger workforce does not necessarily mean faster employee growth.

11.4 Multivariate Analysis

Multivariate analysis examines three or more variables together to discover patterns and relationships in company performance.

1. Correlation Heatmap — Company Performance

The heatmap examines revenue, revenue growth, current workforce and employee growth percentage together.

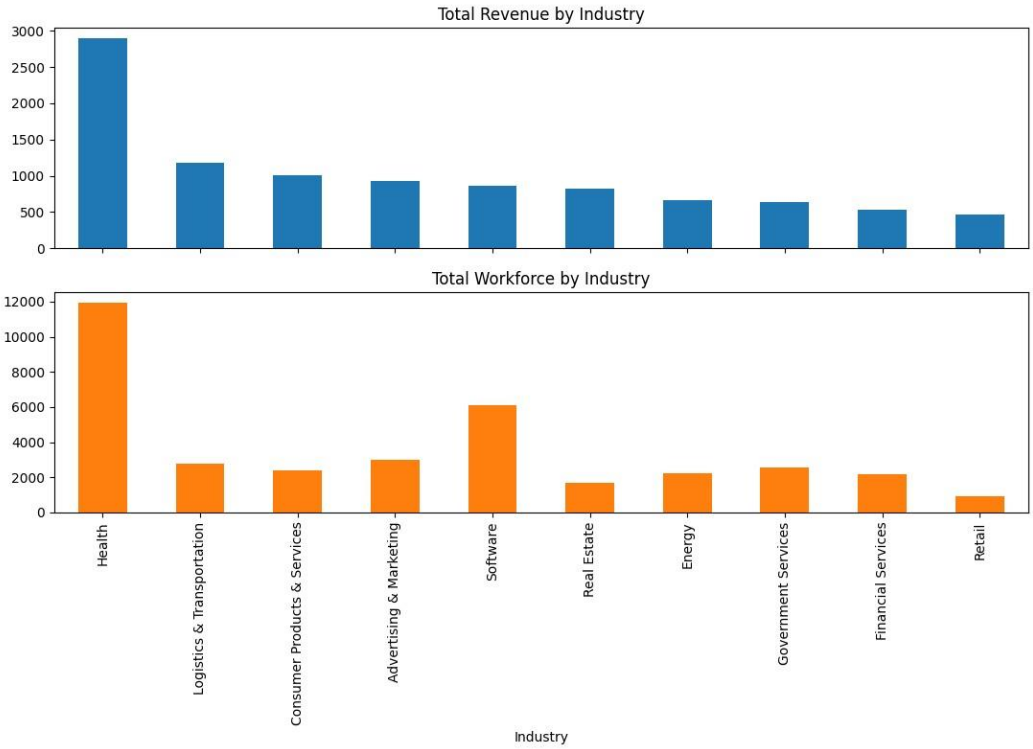


Interpretation: Correlation values range from −1 to +1. Values near zero indicate weak linear relationships. The heatmap supports the earlier finding that revenue scale and workforce size alone do not explain growth performance.

2. Grouped Analysis — Industry Revenue and Workforce

This chart compares total revenue and total current workforce across the ten industries with the highest total revenue.

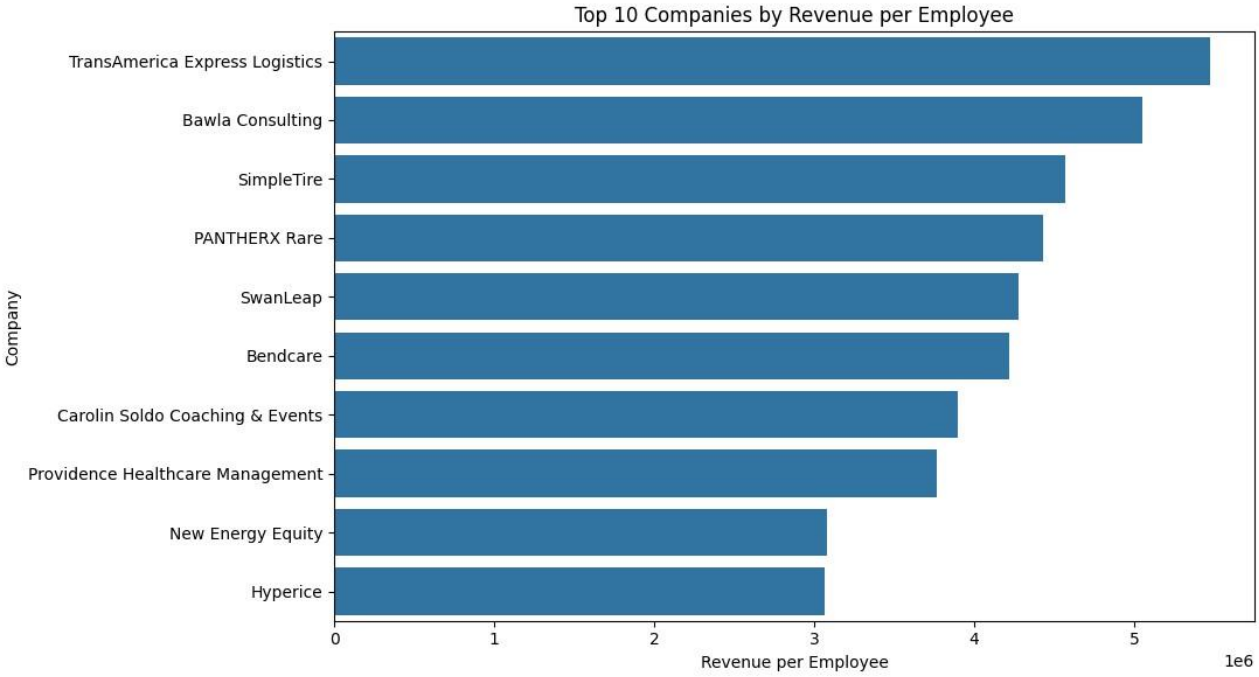
2. Grouped Analysis · Industry Revenue and Workforce



Interpretation: The comparison shows that industries contribute differently to revenue and workforce. High total revenue and a large workforce indicate business scale, but do not by themselves establish profitability or employee productivity.

3. Revenue per Employee — Workforce Productivity

This chart compares revenue generated relative to the current workforce and displays the top ten companies by revenue per employee.



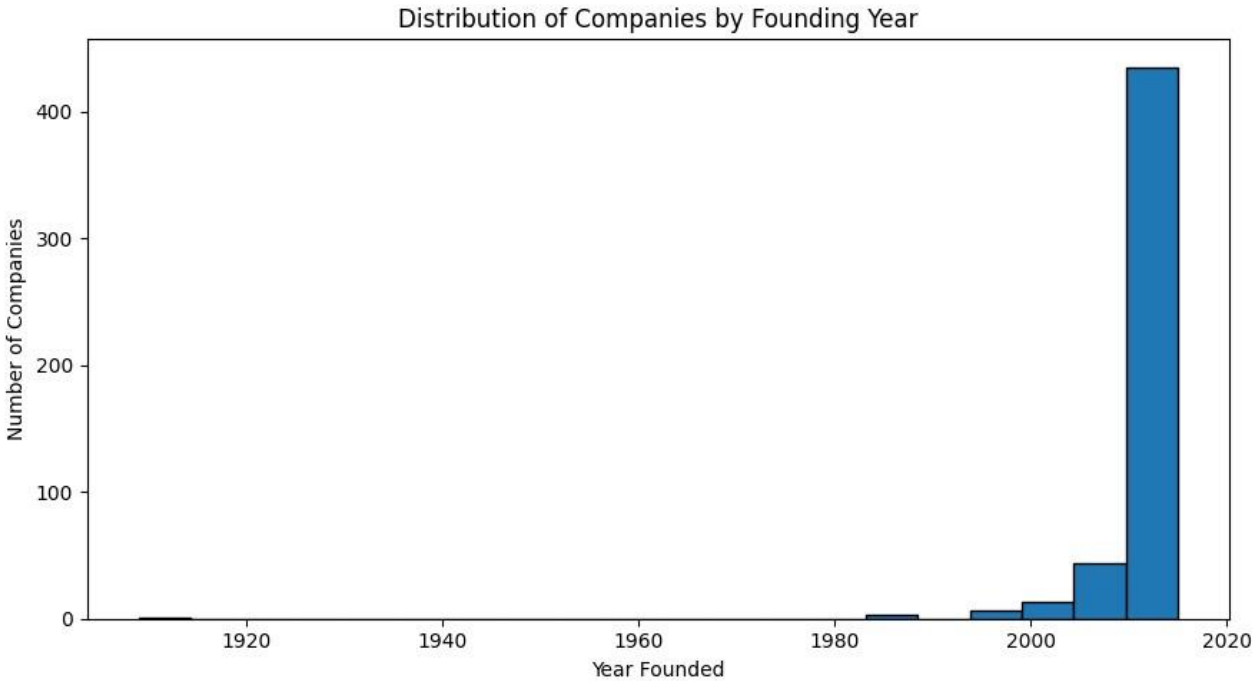
Interpretation: A higher value indicates more revenue relative to the number of current workers. This is a revenue-efficiency measure, not a direct measure of profit or individual employee performance.

11.5 Company Age and Business Trends

This section examines company founding years, compares revenue growth across company age groups and reviews industry representation.

1. Histogram — Distribution of Company Founding Years

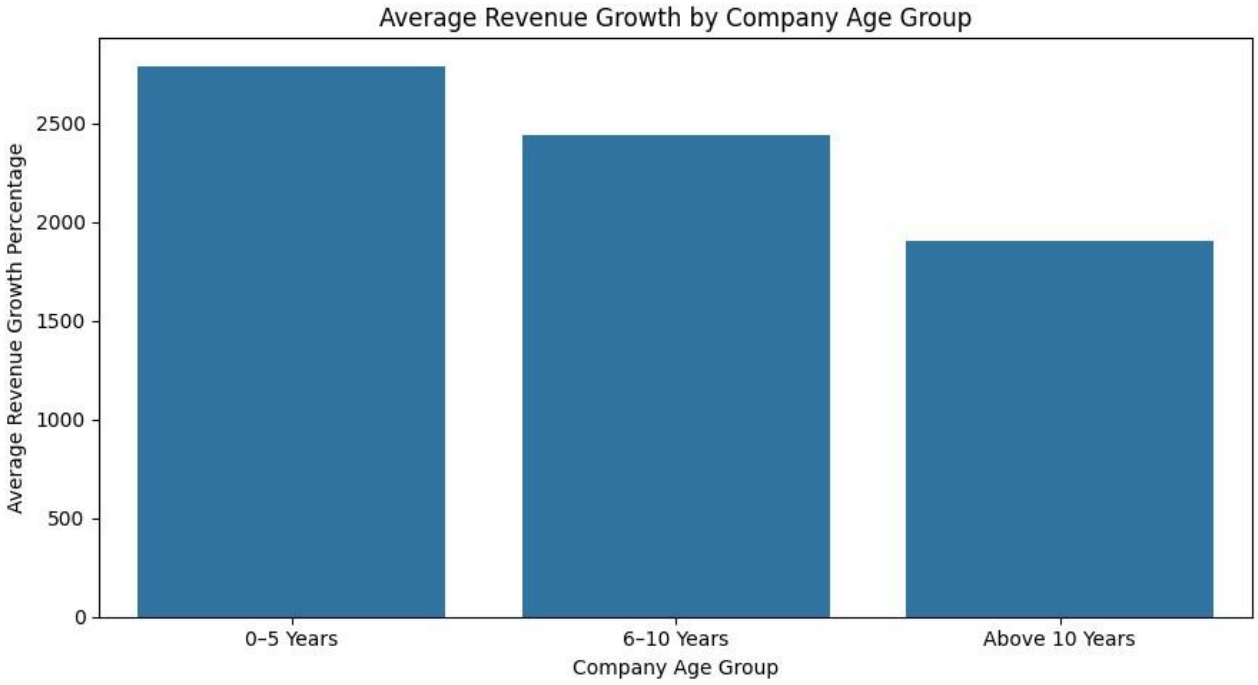
This chart shows when the companies in the dataset were founded.



Interpretation: The distribution highlights the founding periods represented in the sample. It provides context for comparing newer and older companies, but does not by itself show whether newer companies grow faster.

2. Bar Chart — Average Revenue Growth by Company Age Group

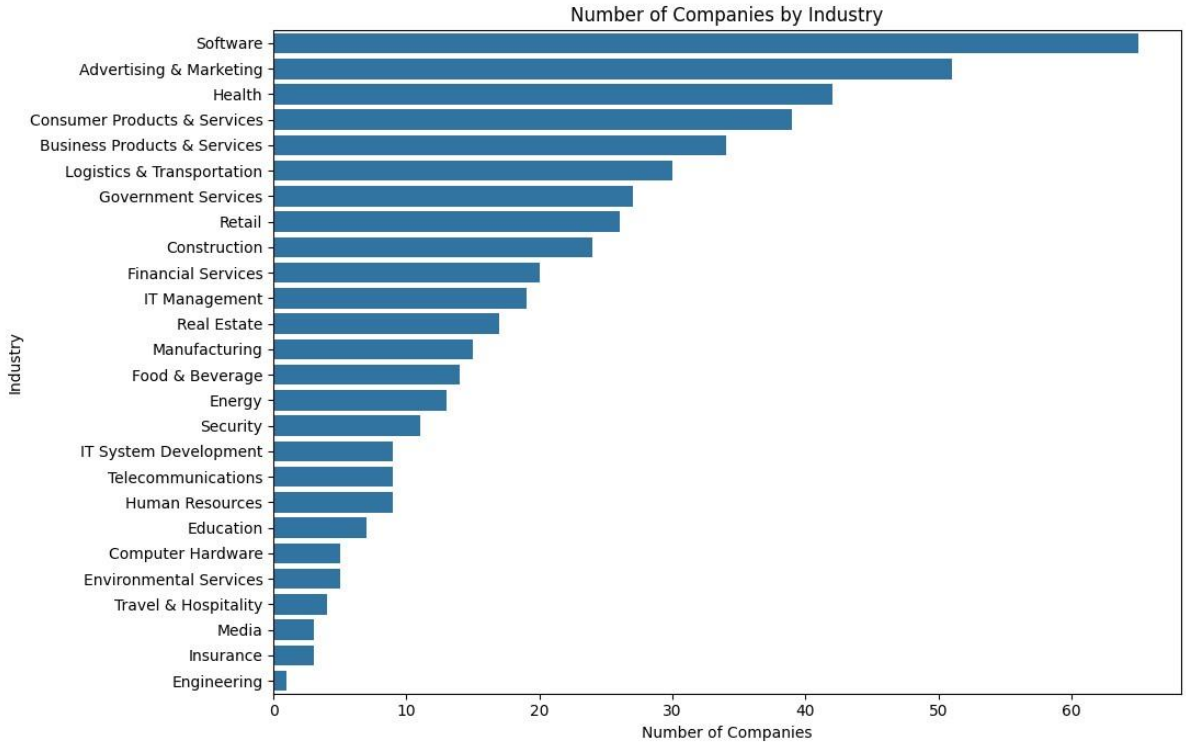
This chart compares average revenue growth for companies aged 0–5 years, 6–10 years and above 10 years.



Interpretation: The comparison helps identify differences in average growth across age groups. Unusually high-growth companies may influence the averages, so median values should also be considered.

3. Bar Chart — Number of Companies by Industry

This chart shows how the 502 companies are distributed across industries.

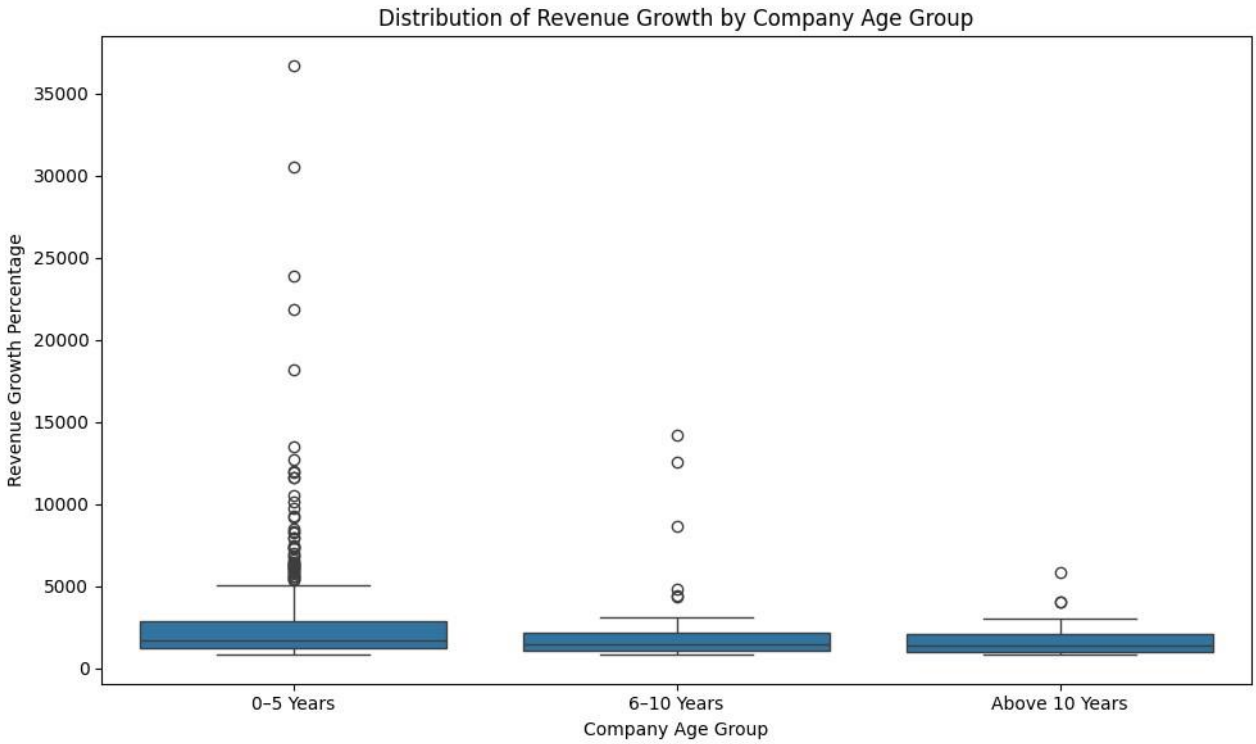


Interpretation: Industries with longer bars have more companies represented in the dataset. A higher company count does not necessarily mean higher revenue or faster growth.

4. Box Plot — Revenue Growth by Company Age Group

This box plot compares the spread and median of revenue growth across company age groups.

4. Box Plot · Revenue Growth by Company Age Group



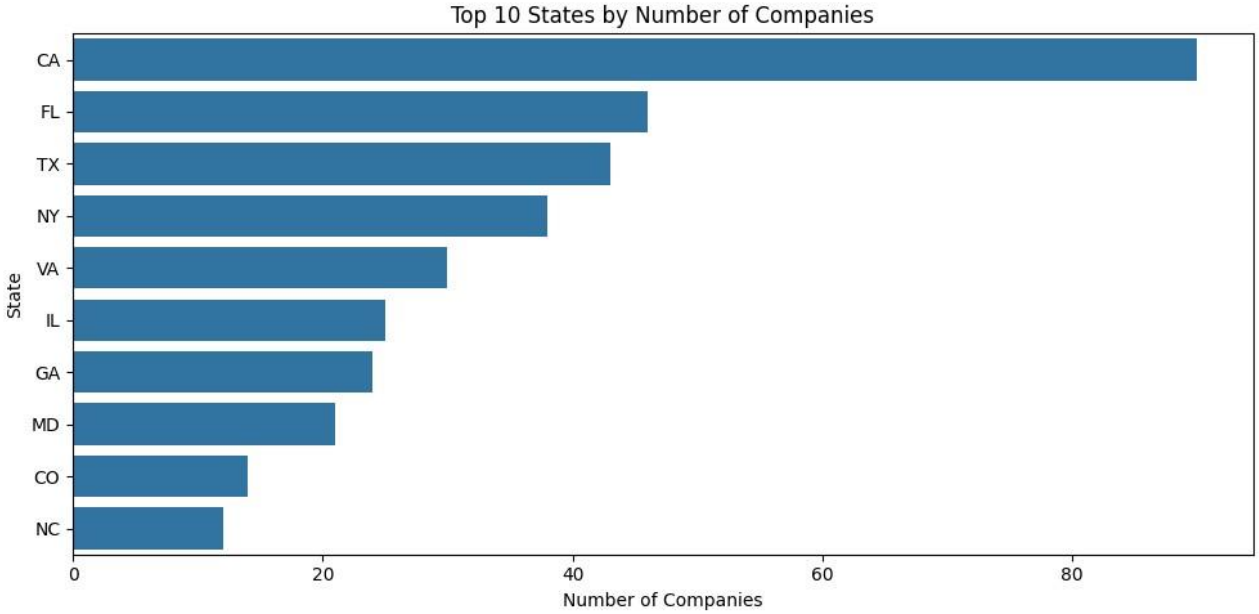
Interpretation: Differences in median lines and box heights indicate whether typical growth and variability differ by company age. Potential outliers should be investigated rather than automatically removed.

11.6 Geographical Analysis

This section compares the number of companies and total company revenue across leading states and cities.

1. Top 10 States by Number of Companies

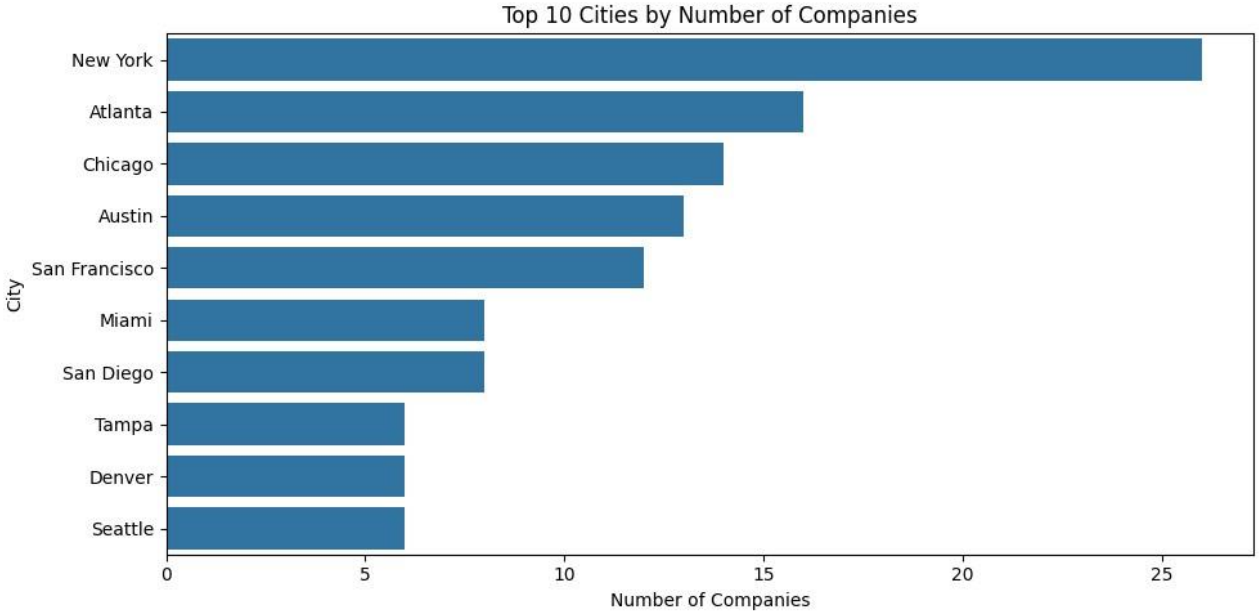
This chart ranks the ten states with the largest company counts.



Interpretation: California leads with approximately 90 companies, followed by Florida with around 46 and Texas with around 43.

2. Top 10 Cities by Number of Companies

This chart ranks the ten cities with the largest company counts.

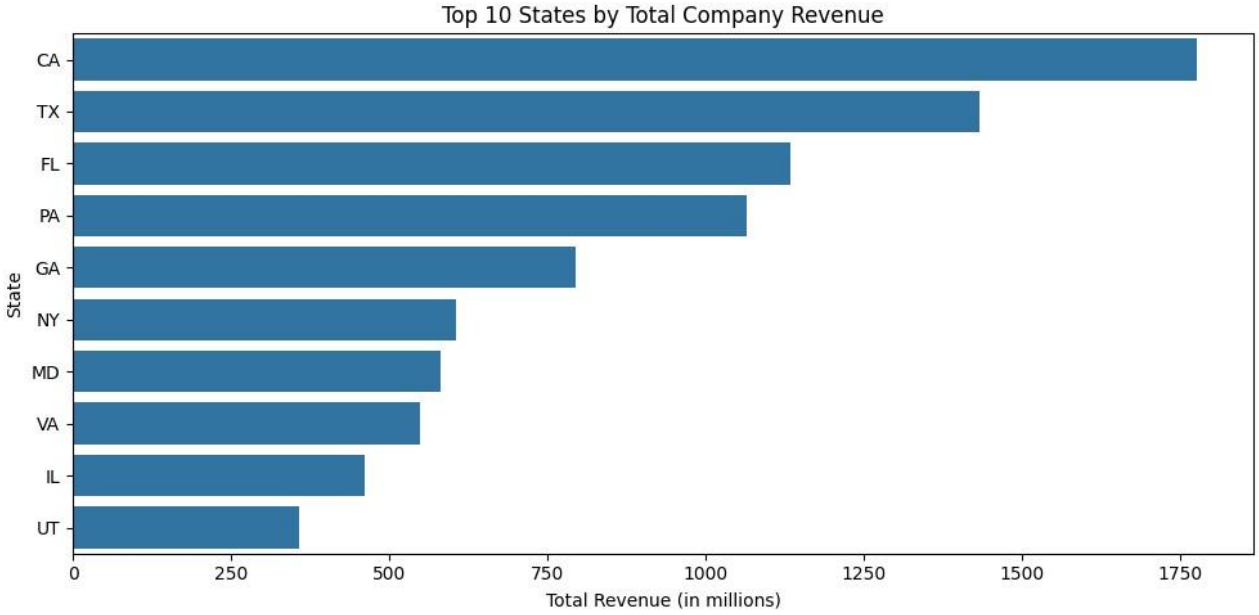


Interpretation: New York leads with approximately 26 companies, followed by Atlanta with around 16, Chicago with 14 and Austin with 13.

3. Top 10 States by Total Company Revenue

This chart compares total company revenue across the ten leading states.

3. Top 10 States by Total Company Revenue



Interpretation: California leads with approximately $1,775 million, followed by Texas at around $1,435 million and Florida at around $1,135 million.

12. Summary of Findings

- Revenue growth differs significantly across companies, with a few companies showing exceptionally high growth percentages.
- The Health industry generates the highest total revenue among the industries analyzed.
- Most companies belong to the Small company category: 298 out of 502 companies.
- California has the highest number of companies among the states analyzed and the highest total revenue among thetop 10 states.
- New York has the highest city-wise company count in the analysis.
- Higher company revenue does not necessarily mean higher revenue growth.
- Larger workforce size does not necessarily indicate faster employee growth.
- Revenue per employee provides a way to compare revenue relative to workforce size.

13. Types of Analysis

1. Descriptive Analysis

Descriptive analysis summarizes historical data and explains what is present in the dataset.

- Calculated mean, median, mode, variance, and standard deviation.
- Analyzed revenue, workforce, company size, industry, and location distributions.
- Created histograms, count plots, box plots, bar charts, scatter plots, and heatmaps.

Outcome: Provided an overall understanding of company growth, revenue, workforce, and business distribution.

2. Diagnostic Analysis

Diagnostic analysis examines relationships and patterns to help understand why certain results may occur.

- Examined the relationship between revenue and revenue growth.
- Compared current workforce size with employee growth percentage.
- Compared revenue growth across company age groups.
- Compared revenue and workforce contributions across industries and locations.

Outcome: The analysis showed that revenue size and workforce size do not necessarily determine higher growth.

3. Predictive Analysis

Predictive analysis uses historical data and statistical or machine learning models to estimate possible future outcomes.

- Future possibility: Predict company revenue using multi-year historical data.
- Future possibility: Estimate workforce growth.
- Future possibility: Apply regression and machine learning models for business forecasting.

Outcome: Predictive models were not developed in the current project; this is a possible future extension.

4. Prescriptive Analysis

Prescriptive analysis suggests actions based on analytical findings.

- Future possibility: Identify industries and locations that may offer business opportunities.
- Future possibility: Suggest workforce planning strategies based on employee growth patterns.
- Future possibility: Use revenue efficiency measures to identify areas for further productivity analysis.

Outcome: A formal optimization or prescriptive model was not developed in the current project. The findings can nevertheless inform business recommendations.

14. Future Enhancements and Suggestions

- Use Updated Datasets: Include more recent company data and compare performance across multiple years.
- Expand the Dataset: Include more companies, industries, locations, profit, expenses, investment, and market-shareinformation.
- Perform Advanced Statistical Analysis: Apply regression and other statistical techniques to study factors associatedwith company growth.
- Apply Machine Learning Techniques: Explore revenue and workforce forecasting and clustering companies withsimilar performance patterns.
- Improve Geographical Analysis: Compare states, cities, and countries and use maps to visualize companyconcentration.
- Develop an Interactive Dashboard: Create a Power BI or Tableau dashboard with filters, charts, and key performanceindicators.
- Conduct Industry-wise Comparisons: Compare revenue, workforce, and growth across industries and company-sizecategories.
- Improve Workforce Productivity Analysis: Add employee costs, department details, and productivity measures whereavailable.
- Handle Outliers and Missing Values More Effectively: Investigate extreme values and improve data-quality checks.
- Perform Time-series Analysis: Use multi-year company data to study long-term revenue and workforce trends.

15. Conclusion

The MNC Growth and Workforce Analytics project used Python to clean, transform, analyze, and visualize the Inc 5000 Companies dataset. The project examined company revenue growth, industry performance, workforce expansion, geographical distribution, revenue per employee, and company age patterns.

The analysis identified important business patterns, including the leading revenue-generating industry, geographical concentration of companies, and weak linear relationships between revenue and revenue growth and between workforce size and employee growth. Data preprocessing, statistical analysis, and visualization helped turn raw company records into meaningful findings.

The results provide a foundation for understanding company performance and supporting data-informed business discussions. Predictive and prescriptive modeling, updated data, and interactive dashboards are possible future enhancements.